

SNEHIL GAUTAM

Mumbai, Maharashtra

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EDUCATION

IIT Jodhpur

2024 – 2028

B.Tech – Artificial Intelligence and Data Science – **CGPA: 6.84**

Jodhpur

TECHNICAL SKILLS

Languages: C, C++, Rust, Python, TypeScript, JavaScript, SQL

Technologies/Frameworks: Node.js, Express.js, React, Next.js, Tauri, WebRTC, WebSockets, Tailwind CSS, Scikit-learn, Rich, Prompt-Toolkit, PyYAML

Tools & Infra: Linux (Arch), Wayland, PipeWire, Docker, Git, Bash/Shell Scripting, PostgreSQL, SQLite, Firebase Admin SDK, Render, Vercel

EXPERIENCE

Research Assistant / Developer

Sept 2025 – April 2026

Department of Computer Science, IIT Jodhpur

Jodhpur, India

- Collaborated under Dr. Bhivraj Suttar to develop an automated Knee Continuous Passive Motion (CPM) medical device integration system designed for personalized orthopedic rehabilitation.
- Engineered the system's control interface and UI wrapper, establishing low-latency state tracking and parameter configurations to safely interface with embedded device hardware.
- Assisted in architecting a predictive ML pipeline framework to dynamically monitor patient pain thresholds, automating real-time adjustments to range-of-motion and speed parameters.

Full-Stack Developer / Participant

Feb 2026

Insomniac Hackathon (PrepClutch) | Github

Remote

- Architected and built an academic analytics platform within 24 hours that ingests unstructured PDFs (lectures, quizzes, PYQs), extracts multi-source raw text via custom parsing scripts, and persists structured data to Firebase.
- Engineered a backend processing service using the Gemini API to analyze textual patterns across historical documents, generating automated, probabilistic topic weight distributions.
- Deployed a responsive Node.js/Express web dashboard featuring restricted uploader access control, a native PDF rendering pane, and dynamic, percentage-based visual trend dashboards.

PROJECTS

Virtual Waynitor | Github | WebRTC, Node.js, WebSockets, Hyprland

April 2026

- Developed a headless virtual monitor for Hyprland streaming directly to browsers using WebRTC and P2P communication without requiring plugins or X11.
- Built a Node.js WebSocket signaling server to relay WebRTC messages, enabling low-latency screen capture via `getDisplayMedia` and PipeWire.

Kyvyrn | Github | React, TypeScript, Tauri, Rust, Vite, Tailwind CSS

Dec 2025

- Built a lightweight, Linux-first desktop application wrapper utilizing Tauri and Rust to convert web applications into native-feeling desktop apps without Electron overhead.
- Implemented smart browser engine detection, per-app isolated configuration, and a responsive React-based interface with Tailwind CSS.

Ani-Loader | Github | Rust, Slint, Btrfs, kexec, QEMU

May 2026

- Engineered a high-fidelity bootloader runtime with Rust and a hardware-accelerated Slint UI, utilizing Linux kexec for seamless kernel handoffs, bypassing slow cold boot hardware initialization.
- Developed native Btrfs subvolume mounting and parsing logic, and established a comprehensive QEMU testing pipeline with mock partitions to emulate production-scale deployments.

ACHIEVEMENTS

- **1st Place / Winner**, Freshmen Electrical & Electronics Engineering Expo (2024): Developed an automated campus attendance prototype using an Arduino microcontroller and an RFID reader module to successfully scan and log existing student ID card data.
- **Academic Distinction**, Principles of Computer Systems (PCS 1): Secured an 'A' grade in core systems engineering coursework covering digital logic design, Instruction Set Architectures (ISA), and fundamental compiler mechanics.